Guidelines for Reporting About Network Data (GRAND)

The Guidelines for Reporting About Network Data (GRAND) are designed to facilitate clarity and transparency in network research reported in scientific documents (e.g., articles, monographs) by providing researchers, editors, and peer reviewers with guidance about the information that should always be reported. In practice, it will often be necessary to report additional information not included in these guidelines depending on the research question and other contextual details.

Please cite GRAND as: Neal, Z. P., Carley, K., Contractor, N., De Domenico, M., Hollway, J., Labianca, G., Landry, N., Lazer, D., Lubbers, M., Neal, J. W., Ortiz, F., Peixoto, T., Pescosolido, B., Prell, C., Teixeira, A. S., and the GRAND Consortium. (2026). Guidelines for Reporting About Network Data. *Pre-print*.

1. Data: The items in this section are intended to provide a description of the network dataset as a whole.

#	Item	Reporting guideline
1.1	**Data**	**If the data are empirical**, report the method originally used to collect them. **If the data are synthetic**, report the data generating model and the values of its parameters.
1.2	**Ethics**	**If the collection or use of these data could present** ethical concerns, report whether institutional or ethical approval was obtained, the approving entity, and approval date.
1.3	**Multiple**	**If the data contain a** panel **of networks representing the same nodes at different times**, report the number of waves and the interval(s) between waves. **If the data contain multiple networks that each represent the same type of system**, report the number of networks.
1.4	**Location**	**If empirical (see 1.1)**, report the location that the data are intended to represent, at a substantively and ethically appropriate level of resolution.
1.5	**Date**	**If empirical (see 1.1)**, report the date(s) that the data are intended to represent, at a substantively and ethically appropriate level of resolution.
1.6	**Availability**	Report the URL or DOI linking to a repository where the data or data-generating code are available, or explain why the data are not available in a repository.
1.7	**Citation**	**If the data have an independent citation**, provide a citation to the data.

2. Nodes: The items in this section are intended to describe the nodes in the network(s).

#	Item	Reporting guideline
2.1	**Nodes**	Report what the nodes are intended to represent. **If the nodes represent more than one type of entity,** describe the network as N-mode and report what each type of node is intended to represent.
2.2	**Size**	Report the number of nodes. **If multiple (see 1.3)**, report the number of nodes in each wave/network if practical; otherwise report the mean, standard deviation, and range. **If multimode (see 2.1)**, report this item separately for each type of node.
2.3	**Boundary**	**If empirical (see 1.1)**, report how the set of nodes that should be included in the network was defined.
2.4	**Missing**	**If nodes that should be included in the network are missing from the network (see 2.3)**, report the percent or number of missing nodes, or the percent or number of present nodes (i.e., the response rate). **If multiple (see 1.3)**, report this value for each wave/network if practical; otherwise report the mean, standard deviation, and range.
2.5	**Degree**	**If nodes' minimum or maximum degree is constrained by the data collection or generation design**, report any restriction(s) on nodes' degree.

3. Edges: The items in this section are intended to describe the edges in the network(s).

#	Item	Reporting guideline
3.1	**Edges**	Report what the edges are intended to represent. **If the edges represent more than one type of relationship**, describe the network as multiplex or multilayer, and report what each type of edge is intended to represent.
3.2	**Direction**	Report whether the edges are directed or undirected. **If directed**, report what the direction of the edge means. **If multiplex (see 3.1)**, report this item separately for each type of edge.
3.3	**Weight**	Report whether the edges are weighted, unweighted, or signed. **If weighted**, report the meaning, scale of measurement, and mean, standard deviation, and range of weights. **If signed**, report the meaning of positive edges and the meaning of negative edges **If multiplex (see 3.1)**, report this item separately for each type of edge.
3.4	**Selfloops**	**If the network contains self-loops**, report that the network contains self-loops and what they represent.
3.5	**Bipartite**	**If multimode (see 2.1),** report where edges can be located.
3.6	**Hypergraph**	**If the network is a hypergraph**, report that the network is a hypergraph.
3.7	**Number**	Report the number of edges present. **If multiple (see 1.3)**, report the number of edges in each wave/network if practical; otherwise report the mean, standard deviation, and range. **If multiplex (see 3.1) or signed (see 3.3)**, report this item separately for each type of edge. **If hypergraph (see 3.6)**, report the number of (hyper)edges, and the mean and standard deviation of the size of the hyperedges.

4. Transformations: The items in this section are intended to describe transformations that were performed on raw data to obtain the final analytic data.

#	Item	Reporting guideline
4.1	Symmetrized	**If directed edges in raw data were transformed into undirected edges in the final analytic network**, report the transformation method and the percent of connected dyads that had non-reciprocal edges before the transformation.
4.2	Dichotomized	**If weighted edges in raw data were transformed into unweighted edges in the final analytic network**, report the transformation method and the number or percent of edges that were deleted as a result of the transformation.
4.3	Projected	**If the final analytic network was generated from two-mode or non-network data such that edges represent similarities, co-occurrences, or statistical associations**, report the transformation method.
4.4	Excluded	**If any nodes or edges that were present in raw data have been excluded from the final analytic network**, report the exclusion criteria and number or percent of nodes or edges that were excluded.
4.5	Aggregated	**If any nodes or edges that were separate in raw data have been aggregated in the final analytic network**, report the aggregation method.
4.6	Imputed	**If any nodes or edges were unmeasured in raw data, and their presence or absence was imputed or inferred in the final analytic network**, report the imputation method and the number or percent of nodes or edges that were imputed.